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DAWN: Dimension-aware graph contrastive learning for few-shot dissolved gas analysis

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ABSTRACT

Dissolved Gas Analysis (DGA) is a widely used technique for identifying characteristic gas signatures indicative of transformer faults. However, traditional DGA methods struggle to capture the complex interdependencies among gases. Although deep learning has shown promise in this domain, existing approaches face significant limitations under few-shot learning scenarios, primarily due to severe class imbalance and high inter-class similarity, which lead to diagnostic ambiguity. Moreover, these methods often overlook critical inter-gas relationships that are essential for understanding underlying fault mechanisms. To address these challenges, we propose **DAWN** (**D**imension-**AW**are Graph Co**N**trastive Learning), a novel framework that integrates two key components: (1) a contrastive few-shot learning module with clustering consistency loss to enhance the discriminability of similar fault categories through lightweight fine-tuning, and (2) a knowledge-enhanced dimension graph that explicitly models structural dependencies among gas features by combining statistical correlations with expert domain knowledge. Extensive evaluations on DGA datasets demonstrate that DAWN achieves state-of-the-art performance, improving rare fault detection accuracy by over 15% compared to conventional methods. To the best of our knowledge, this work represents the first contrastive few-shot learning framework tailored for DGA-based fault diagnosis.

1. Introduction

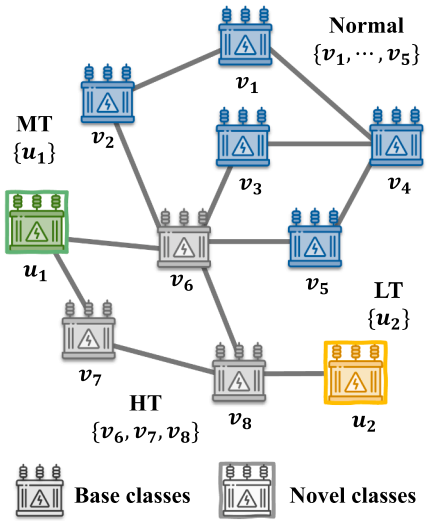
Power transformers play a pivotal role in contemporary power systems, acting as a crucial conduit connecting generation, transmission, and distribution networks. Their operational stability stands as a linchpin, directly impacting grid safety, reliability, and efficiency (Balanta et al., 2023). However, prolonged exposure to high loads, voltage stresses, and adverse environmental conditions predisposes transformers to an array of faults, including winding shorts, localized core overheating, insulation degradation, and abnormal gas generation within the insulating oil. These faults not only hasten equipment aging but also have the potential to trigger severe consequences such as grid outages, large-scale blackouts, or even catastrophic failures, underscoring the critical necessity for intelligent fault diagnosis.

Dissolved Gas Analysis (DGA) has emerged as the predominant diagnostic technique for early transformer fault detection. When internal faults manifest, insulating materials decompose, generating characteristic gases (H_2, CH_4, C_2H_6, C_2H_4, C_2H_2). The concentrations and ratios of these gases serve as indicators of specific fault types. While traditional DGA methods (Ali et al., 2023; Li, 2024) typically rely on empirical thresholds and established ratio rules (e.g., IEC 60599), they often fall short in effectively capturing the intricate inter-gas relationships, thereby limiting their adaptability to novel or compound fault scenarios. Deep learning approaches (Cao et al., 2024; Zemouri, 2025) present a way to address these limitations by extracting hidden patterns directly from gas concentration data (Fanchiang et al., 2021). Nevertheless, many existing methods predominantly focus on individual sample features, overlooking the inherent structural relationships between samples. Recent advancements in graph-based methods (Shu et al., 2024; Zhang et al., 2025) have tackled this issue by constructing sample graphs, where nodes represent oil samples and edges signify feature similarities, and leveraging Graph Neural Networks (GNNs) (Kipf & Welling, 2017), which have shown enhanced diagnostic performance.

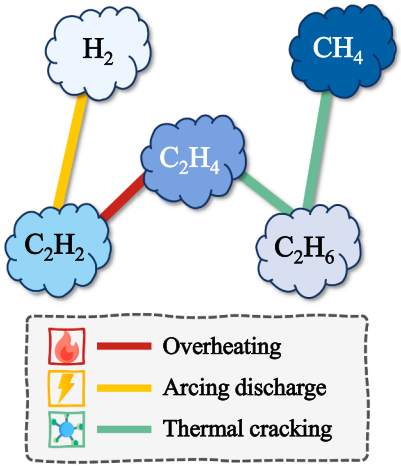
Despite their effectiveness, two critical challenges persist within this domain. **Firstly, few-shot learning limitations with inter-class confusion.** Under extreme data scarcity, GNNs struggle to bridge the distribution gap between training-phase categories (common faults) and test-phase categories (rare faults). As shown in Fig. 1a, while common

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(a) Few-shot node classification with base and novel classes in the context of DGA.



(b) Inter-dimensional correlations: Latent structural relationships reflect physical conversion mechanisms including thermal cracking, electrical discharge and overheating.

Fig. 1. The graph-based framework for few-shot node classification in DGA, alongside the latent structural relationships between gases. Base classes contain abundant labeled nodes (e.g., Normal, HT), while novel classes have only a few labeled nodes (e.g., LT, MT). Latent structural relationships among gas feature dimensions are reflected in their interconversion mechanisms, including thermal cracking, electrical discharge, and overheating.

faults (e.g., Normal, High Thermal) have abundant samples, rare faults (e.g., Low/Medium Thermal) are severely underrepresented. Simultaneously, substantial semantic overlap exists between certain fault categories (e.g., Medium/High Thermal; Partial/High-Energy Discharge) due to similar gas compositions, resulting in ambiguous class boundaries. Data scarcity and inter-class similarity hinder discriminative representation learning, compromising model generalization and few-shot diagnostic accuracy, making pretrain-finetune strategies (Tan et al., 2022) and discriminative loss functions crucial for effective fault detection. **Secondly, inter-gas relationship oversight.** Existing methods frequently neglect correlations and synergistic effects between gas dimensions (e.g., C_2H_2-C_2H_4 interactions) as suggested in Fig. 1b. Indeed, complex feature relationships exist between different gas dimensions (detailed analysis are provided in Section 3), yet current models fail to leverage this information. Explicitly modeling the interdependencies between gas dimensions could capture latent synergies, thereby enhancing fault representation quality and diagnostic accuracy.

In response to these challenges, this paper introduces **DAWN**, a novel **D**imension-**AW**are Graph Co**N**trastive Learning method tailored for few-shot dissolved gas analysis. To overcome the first limitation, we develop **a contrastive few-shot learning framework.** Initially, we leverage similarity and dissimilarity sample graphs to pretrain a graph encoder. Subsequently, we incorporate a novel clustering consistency contrastive loss, designed to boost discriminative capabilities, especially for distinguishing between high-similarity fault classes. This facilitates effective and lightweight fine-tuning even when only minimal support samples are available. To address the second limitation, we propose a **knowledge-enhanced dimension graph** that explicitly models inter-gas relationships from two aspects. For one thing, edges are constructed based on statistical correlations between gas dimensions. For another, the graph topology is refined by integrating domain knowledge, exemplified by gas ratios. The derived dimension graph embeddings enhance sample representations by encoding diagnostically critical gas synergies.

To summarize, the main contributions of our work are as follows:

- **Innovative contrastive few-Shot diagnosis method.** We introduce a novel contrastive few-shot diagnosis method specifically tailored for DGA. This method incorporates a clustering consistency contrastive loss to boost discriminative power, especially for fault classes with high similarity. It enables efficient and lightweight fine-tuning even with limited support samples.
- **Pioneering dimension-graph framework with data and domain knowledge integration.** We present the first dimension-graph framework crafted for fault diagnosis, seamlessly merging data-driven statistical correlations among gas dimensions with domain-specific physical knowledge. This integration refines the graph topology through two stages, resulting in more informative dimension graph embeddings.
- **Competitive performance on real-world data.** Through extensive experiments on actual DGA datasets, we validate the efficacy and resilience of our proposed methodologies. Our results demonstrate substantial accuracy improvements, notably achieving over a 15% enhancement in rare fault detection compared to conventional methods.

The structure of this paper is as follows: Section 2 reviews related work; Section 3 formally defines the few-shot learning problem within this context; Section 4 details our proposed methodology; Section 5 presents the experimental setup, results, and analysis; finally, Section 6 concludes the paper and outlines potential future research directions.

2. Related works

2.1. Transformer fault diagnosis

Transformers play a vital role in modern power systems, and DGA has become a widely used and effective technique for fault diagnosis. Traditional DGA methods, such as the Duval Triangle, Key Gas Method and IEC 60,599 Ratio Method (Abbasi, 2022), are simple to apply but rely heavily on empirical rules. This makes them less adaptable to complex or evolving fault patterns, often leading to poor generalization and limited diagnostic accuracy under non-standard operating conditions.

To address these limitations, data-driven machine learning approaches have been widely adopted in scientific, industrial, and other application domains (Fu et al., 2024; Yu et al., 2020, 2025). In the context of transformer fault diagnosis using dissolved gas analysis (DGA),

for example, methods such as Support Vector Machines (SVM) (Hong et al., 2022; Zhang et al., 2017) and Random Forests (RF) (Prasojo et al., 2023) have been introduced to learn fault patterns directly from data. However, these models typically treat samples independently and overlook structural dependencies, which limits their effectiveness when data is scarce or imbalanced.

Recently, GNNs have been applied to DGA tasks (Shu et al., 2024; Zhang et al., 2025), enabling the modeling of sample relationships through graph structures, where nodes represent data instances or features and edges encode potential dependencies. Furthermore, leveraging multi-view graph structures-such as high-pass views that capture local variations and low-pass views that reflect global smoothness-can further enhance the model's ability to identify diverse fault types (Jin et al., 2024). Compared to traditional techniques, these graph-based methods offer more robust and adaptable diagnostic capabilities.

2.2. Few-shot node classification

Despite the success of graph-based models in DGA fault diagnosis, real-world applications often face a few-shot scenario, where labeled data is extremely limited. Due to the high cost of expert annotations and the rarity of certain fault types, developing effective models under such constraints remains a pressing challenge.

To address the few-shot node classification problem on graphs, **meta-learning** has emerged as an effective solution (Yu et al., 2024). Meta-learning aims to extract transferable knowledge from multiple tasks so that models can quickly adapt to new tasks with limited data (Vettoruzzo et al., 2023). In the graph domain, many approaches integrate GNNs with meta-learning frameworks, such as Model-Agnostic Meta-Learning (MAML) (Zhou et al., 2019), to construct task-specific adaptations. These methods often leverage local subgraph structures and shared embeddings to enable rapid adaptation and achieve better performance on new, unseen classes. However, meta-learning methods also have certain limitations: first, their training process relies on a large number of tasks and requires careful task design, otherwise generalization may be insufficient; second, when graph structures are complex or class distributions are highly imbalanced, models can be sensitive to noisy nodes or extreme samples, leading to performance fluctuations.

Meanwhile, **Graph Contrastive Learning (GCL)** has also become an important direction for few-shot scenarios (Luo et al., 2021; Tan et al., 2022). Unlike meta-learning, contrastive learning focuses on learning a discriminative encoder by contrasting positive and negative sample pairs (Yang et al., 2022). Once trained, the encoder can be transferred to downstream tasks, where only a small number of labeled samples are required for fine-tuning (Jaiswal et al., 2021; Tian et al., 2022). GCL-based approaches offer strong representation ability, simplicity, and high transferability, making them well-suited for label-scarce environments. However, GCL also has limitations: during the construction of positive and negative pairs, the method is sensitive to noise and intra-class variations; if negative samples are not properly selected, it may lead to unstable representations or insufficient inter-class discrimination.

In summary, meta-learning methods emphasize rapid adaptation through task-level optimization, while contrastive learning methods aim to learn transferable representations that generalize well across tasks (Zhang et al., 2022). Both approaches offer complementary strategies for addressing few-shot node classification problems. Nevertheless, in practical scenarios such as DGA fault diagnosis, how to fully exploit graph structural information while overcoming data scarcity and class imbalance remains an open and challenging problem.

3. Preliminaries

In this section, we first introduce the relevant notations and formally define the few-shot node classification problem in the context of DGA. We then analyze the characteristics of two publicly accessible DGA benchmark datasets- **DGA-2000** [1] and **DGA-200** [2] (detailed statistical information are summarized in Section 5), focusing on key challenges. This provides a foundation for the subsequent method development and experimental studies.

3.1. Notations and problem definition

In this study, we provide a formal definition of the notation and the few-shot node classification problem in the context of transformer fault diagnosis.

Notations. Given the original oil sample feature matrix $\mathbf{X} \in \mathbb{R}^{N\times d}$, where N denotes the number of samples and d denotes the number of measured gas types. Each row $\mathbf{x}_i \in \mathbb{R}^d$ corresponds to the feature vector of the ith sample, with each dimension representing the concentration of a specific gas. We then construct two undirected attributed graphs: the similar low-pass graph $G^l = (V, E^l, \mathbf{A}^l, \mathbf{X})$ and the dissimilar high-pass graph $G^h = (V, E^h, \mathbf{A}^h, \mathbf{X})$. Both graphs share the same node feature matrix $\mathbf{X}$, while their adjacency matrices $\mathbf{A}^l$ and $\mathbf{A}^h$ differ and are constructed based on *K-Nearest Neighbors* (*KNN*) in the feature space to capture distinct structural relationships, with $\mathbf{A}^l, \mathbf{A}^h \in \{0,1\}^{N\times N}$. The node set $V = \{v_1, \ldots, v_N\}$ contains all samples, and the edge sets E^l and E^h represent the connections in the high-pass and low-pass graphs, respectively. Furthermore, for each node v_i, a label mapping function $f : v_i \mapsto y_i$ is defined, where $y_i \in C$ denotes the fault class label of node v_i, and C is the set of all classes, containing $|C|$ distinct fault categories.

Few-shot node classification task. Based on the above graph structure, we define the few-shot node classification task for DGA. The class label space C is partitioned into disjoint subsets for training, validation, and testing, denoted as C_{train}, C_{val}, and C_{test}, respectively, with the constraint $C_{\text{train}} \cap C_{\text{test}} = \emptyset$. Few-shot learning typically follows an n-way k-shot setting, where in each task, the support set S contains k labeled samples per class from n novel classes, and the query set Q contains unlabeled samples to be classified. The objective is to classify the samples in Q based on the limited labeled information in S.

The goal is to train a graph encoder that can transfer the topological and semantic knowledge learned from the base classes C_{train} to the novel classes C_{test}, generating discriminative node embeddings to enable accurate classification with only limited labeled data.

3.2. Design motivation

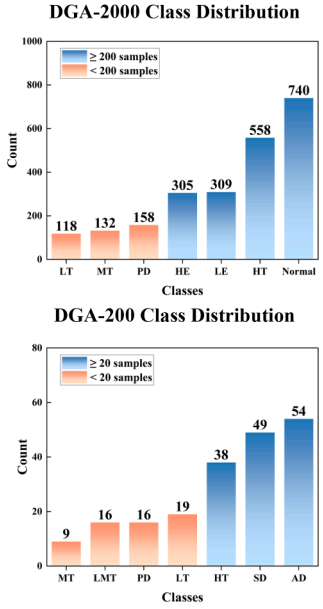
To better understand the structural characteristics of the DGA dataset and provide empirical motivation for our method design, we conduct a systematic data-driven analysison. Specifically, we perform in-depth analyses from two key perspectives: (1) class-level data characteristics, including sample quantity and inter-class similarity, and (2) the intrinsic interdependencies among gas features. These analyses not only validate the identified challenges but also inform the guiding principles behind the design of our proposed approach.

3.2.1. Class distribution and inter-class similarity analysis

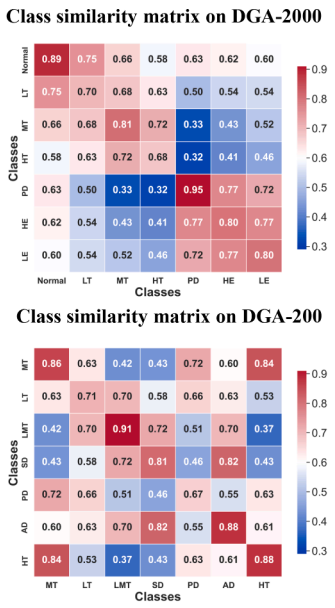
Both DGA datasets exhibit significant class imbalance across fault categories, with rare fault types like Low Thermal (LT) and Medium Thermal (MT) comprising $< 10\%$ of samples in both DGA-2000 and DGA-200 (Fig. 2a). This distribution reflects real-world scarcity of critical faults and associated annotation challenges, motivating our few-shot node classification framework. Critically, substantial semantic similarities exist between fault categories - evidenced by high average cosine similarity scores between PD-HE (0.77) and MT-HT (0.84) pairs

[1] https://github.com/alan-456/transformer-fault-dataset

[2] https://ieee-dataport.org/documents/dissolved-gas-data-transformer-oilfault-diagnosis-power-transformers-membership-degree



(a) Class distribution in the DGA-2000 and DGA-200 datasets.



(b) Inter-class similarity matrix computed using average cosine similarity.

Fig. 2. Sample distribution across fault categories in the DGA-2000 and DGA-200 datasets, alongside the inter-class similarity matrix computed using average cosine similarity. Several fault categories, including LT and MT, account for less than 10% of the total samples. The similarity matrix indicates strong semantic similarity between PD and HE on DGA-2000, and between HT and MT on DGA-200, exhibiting high similarity scores of 0.77 and 0.84, respectively.

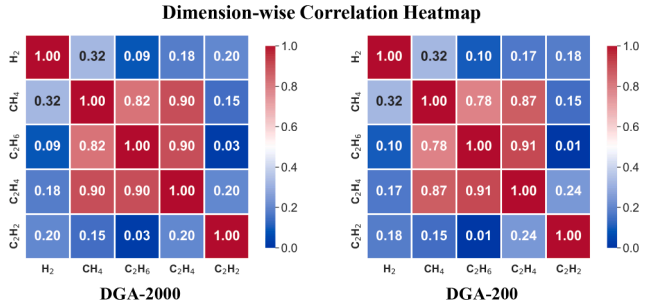


Fig. 3. Pearson correlation heatmap of the 5 gas feature dimensions in the DGA-2000 and DGA-200 datasets. The pairs CH_4–C_2H_4 and C_2H_4–C_2H_2 show high Pearson correlation coefficients of both 0.90 in DGA-2000, and 0.87 and 0.91 in DGA-200.

(Fig. 2b). These overlaps stem from shared gas compositions, creating ambiguous inter-class boundaries that exacerbate representation learning difficulties under low-data regimes.

3.2.2. *Inter-dimensional correlation analysis*

By computing Pearson correlation coefficients across all gas pairs and visualizing the results via a correlation heatmap (Fig. 3), we quantify inter-gas dependencies, uncovering two critical patterns: (1) Strong positive correlations between CH_4–C_2H_4 ($r = 0.90/0.87$) and C_2H_4–C_2H_6 ($r = 0.90/0.91$) in both datasets, signifying shared thermal decomposition pathways; and (2) Weak correlations in H_2–C_2H_2 ($r < 0.3$), reflecting independent generation mechanisms. These statistically significant dependencies prove that gas features exhibit latent structural couplings, not acting independently as often assumed. Actually, prevailing methods commit dual modeling oversights-feature isolation (treating gas dimensions as orthogonal inputs) and relationship neglect (failing to incorporate covariance matrices)-which inherently limits their ability to capture synergistic fault mechanisms, ultimately leading to suboptimal feature expressiveness in downstream tasks.

To sum up, few-shot fault diagnosis for DGA faces two main challenges: (1) limited fault samples and high inter-class similarity, and (2) insufficient modeling of the latent structural dependencies among gas features. To address these issues, we propose a dual-path solution: (1) enhancing class separability through clustering-consistency contrastive learning, and (2) capturing inter-feature dependencies via dimension-level relational graph modeling. The following sections provide detailed descriptions of these two core modules.

4. Methodology

This section introduces DAWN, a framework that learns discriminative node representations from limited labeled data to enhance novel fault type recognition for DGA. Unlike single-view methods, DAWN models both structural sample relationships and latent dependencies among gas feature dimensions, forming a comprehensive multi-view graph representation, combining pretraining and fine-tuning for generalization. The framework (Fig. 4) comprises four modules. Firstly, **(a) Graph construction** builds three graphs from gas features: a sample-level high-pass graph (capturing differences), a low-pass graph (capturing similarities), and a dimension-level graph (modeling feature dependencies), providing multi-level structural support. Next, **(b) Pretraining & dimension-aware enhancement** module optimizes the encoder via contrastive learning on multi-view graphs, distinguishing positive/negative pairs. It integrates dimension-level embeddings to enrich sample representations with feature dependency information. Finally, **(c) Fine-tuning & classification** module adapts the model to new fault categories using few labeled samples, fine-tuning a lightweight classifier for unseen type recognition. Moreover, **(d) Clustering-Consistency contrastive loss** module complements pretraining by adaptively weighting negative samples to mitigate inter-class similarity challenges, improving separability and discriminative performance.

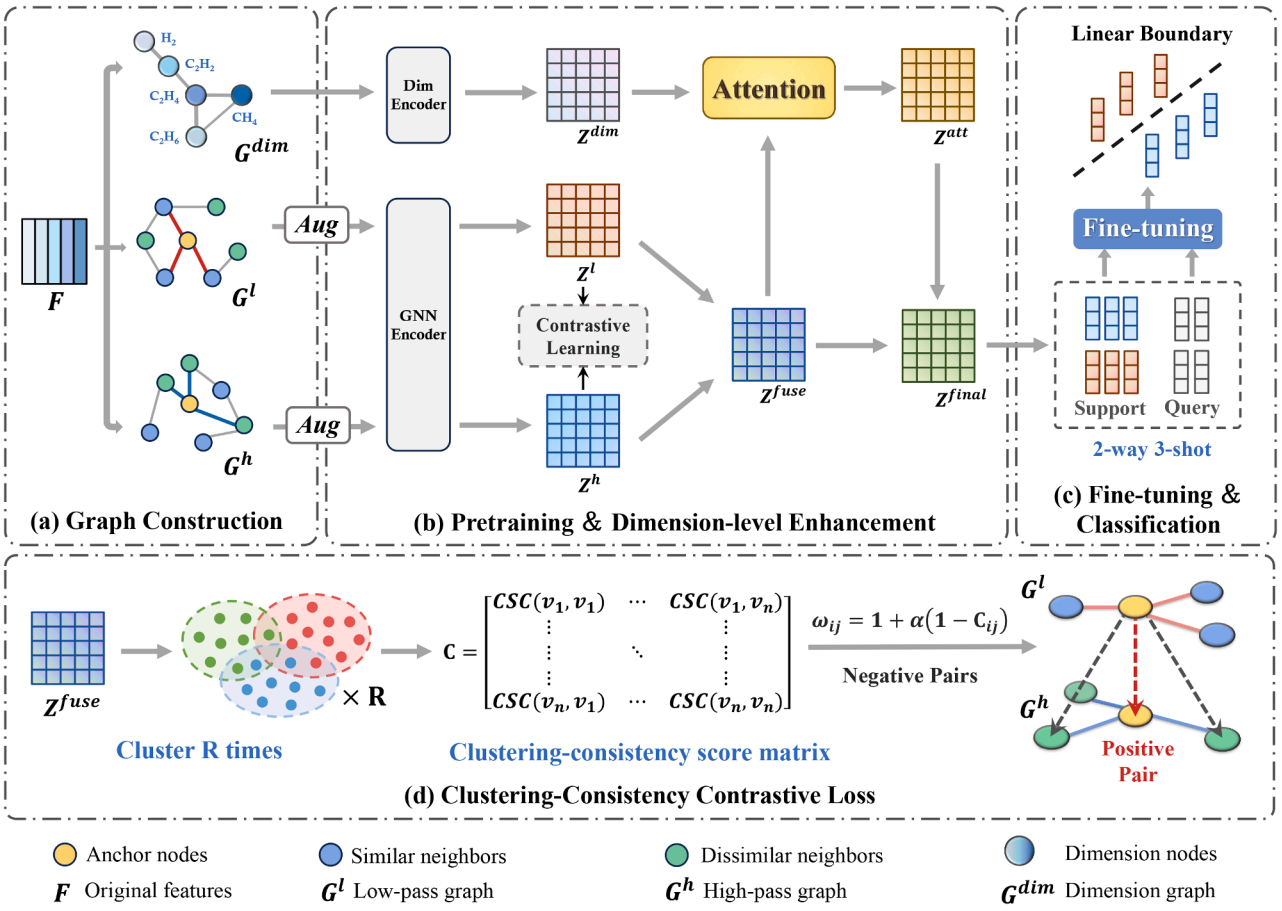


Fig. 4. The presented framework, illustrated through its four core stages: (a) Graph Construction uses raw features to build complementary sample- and dimension-level graphs. (b) Pretraining & Enhancement involves contrastive learning for graph encoders, further refined using dimension graph embeddings. (c) Fine-tuning & Classification adapts a linear classifier for few-shot learning using limited labeled samples from novel classes. (d) The Clustering-Consistency Contrastive Loss, applied during pretraining, reduces inter-class similarity to enhance feature discriminability and class separability.

4.1. Graph construction

4.1.1. Sample graph construction

Given that the original DGA data is discrete and lacks explicit structural information, we represent each sample as a node in a graph to uncover latent relationships. Following previous studies (Shu et al., 2024; Zhang et al., 2025), we construct a low-pass graph and a high-pass graph. We compute cosine similarity between samples and apply the *K-Nearest Neighbors* (KNN) algorithm (Sewwandi et al., 2023) to establish connections. Each node is connected to its top-k most and least similar neighbors, forming the low-pass and high-pass graphs, respectively (see Fig. 5). Formally, these two graphs are defined as:

$$\mathbf{G}^l = (\mathbf{A}^l, \mathbf{X}), \quad \mathbf{G}^h = (\mathbf{A}^h, \mathbf{X}), \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{N\times d}$ denotes the feature matrix of all nodes, and $\mathbf{A}^l, \mathbf{A}^h \in \mathbb{R}^{N\times N}$ are the adjacency matrices of the low-pass and high-pass graphs, respectively.

This dual-view graph structure provides complementary information for contrastive learning, enabling the model to learn robust and discriminative representations by considering both intra-class consistency and inter-class separability, thereby improving performance on few-shot node classification tasks.

4.1.2. Dimension graph construction

In domains such as transformer fault diagnosis, feature dimensions often carry clear semantic meanings. To effectively capture the latent relationships among these dimensions, we introduce a dimension graph. As illustrated in Fig. 5, the dimension graph is constructed by integrating two sources of information: (1) statistical correlations between feature dimensions, which define the initial edge structure; and (2) prior domain knowledge, such as known gas ratios, which is used to refine and enhance graph connectivity.The following section details the construction process of the dimension graph.

With the input feature matrix $\mathbf{X}$, the construction process begins by standardizing each feature to have zero mean and unit variance:

$$\tilde{\mathbf{X}} = \text{Standardize}(\mathbf{X}). \tag{2}$$

Next, we compute a correlation matrix $\mathbf{S} \in \mathbb{R}^{d\times d}$ to capture the linear relationship between any pair of feature dimensions i and j:

$$\mathbf{S}_{ij} = \frac{1}{N}\sum_{k=1}^{N} \tilde{\mathbf{X}}_{k,i} \cdot \tilde{\mathbf{X}}_{k,j}, \tag{3}$$

where $\mathbf{S}_{ij}$ denotes the correlation between dimension i and dimension j, and $\tilde{\mathbf{X}}_{k,i}$ is the normalized value of the ith feature for the kth sample, with the resulting correlation matrix $\mathbf{S}$ serving as a basis for constructing the dimension graph and facilitating the discovery of latent dependencies among feature dimensions.

To obtain a sparse graph structure, we apply a threshold γ and retain only those pairs with sufficiently strong correlations:

$$\mathbf{A}^{\text{dim}}_{ij} = \begin{cases} |\mathbf{S}_{ij}|, & \text{if } |\mathbf{S}_{ij}| > \gamma \text{ and } i \neq j \\ 0, & \text{otherwise} \end{cases}. \tag{4}$$

Here, $\mathbf{A}^{\text{dim}}_{ij}$ is the element of the adjacency matrix $\mathbf{A}^{\text{dim}} \in \mathbb{R}^{d\times d}$, and γ is a correlation threshold that controls the sparsity of the graph. By selecting

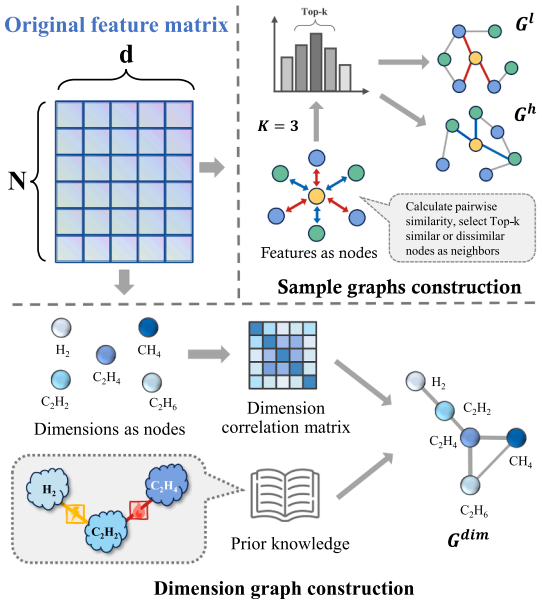


Fig. 5. Illustration of the multi-view graph construction process. The sample-level graphs treat samples as nodes and construct the low-pass and high-pass graphs based on similarity and dissimilarity relationships, respectively. The dimension graph treats feature dimensions as nodes and establishes edges by integrating correlation between dimensions and prior knowledge.

appropriate γ, we ensure that only strongly correlated feature pairs are connected in the dimension graph.

In addition to the data-driven correlations, we incorporate **prior knowledge** to further refine the graph structure in alignment with real-world scenarios (Nanfak et al., 2021). Specifically, we manually add edges between feature pairs that are known to interact through physical or chemical mechanisms, assigning them relatively high weights to emphasize their importance. For example:

- H_2 and C_2H_2 (arcing discharge),
- C_2H_6 and C_2H_4 (overheating),
- C_2H_4 and C_2H_2 (thermal cracking).

Finally, we represent the weighted dimension graph as $\mathbf{G}^{\text{dim}} = (\mathbf{A}^{\text{dim}}, \mathbf{X}^T)$, where $\mathbf{A}^{\text{dim}}$ is the weighted adjacency matrix of the dimension graph, and $\mathbf{X}^T \in \mathbb{R}^{d\times N}$ is the transpose of the sample feature matrix $\mathbf{X}$. By learning representations from the dimension graph $\mathbf{G}^{\text{dim}}$ and incorporating them into the sample-level embeddings, the model is better equipped to capture complex inter-feature interactions, thereby improving performance in few-shot learning scenarios.

In total, we construct three complementary graphs: a low-pass graph $\mathbf{G}^l$, a high-pass graph $\mathbf{G}^h$, and a dimension graph $\mathbf{G}^{\text{dim}}$, each capturing distinct aspects of the data. The first two graphs model sample-level structural relationships, while the dimension graph captures inter-feature dependencies to enhance the learning of sample representations in the next stage.

4.2. Pretraining and dimension-level enhancement

4.2.1. Pretraining the graph encoder

After constructing the multi-view graphs, our core objective is to pretrain the GNN encoder by leveraging these multi-view graph structures, thereby learning discriminative and robust node representations that lay a solid feature foundation for downstream tasks. To this end, we design a staged and differentiated encoding scheme tailored for different graph types.

First, for the sample-level low-pass and high-pass graphs, we perform augmentations, including edge dropping and feature masking (You et al., 2021; Zhao et al., 2020), to obtain augmented graphs $\tilde{\mathbf{G}}^l = (\tilde{\mathbf{A}}^l, \tilde{\mathbf{X}}^l)$ and $\tilde{\mathbf{G}}^h = (\tilde{\mathbf{A}}^h, \tilde{\mathbf{X}}^h)$, which are then used in the pretraining phase.

Let $\mathcal{D}_{\text{pre}}$ denote the set of all multi-view augmented sample pairs used in pretraining. The encoder parameters θ are obtained by minimizing the pretraining loss $\mathcal{L}_{\text{pretrain}}$:

$$\theta = \arg\min_{\theta} \mathcal{L}_{\text{pretrain}}(\mathcal{D}_{\text{pre}}; \theta). \tag{5}$$

Here, $\mathcal{L}_{\text{pretrain}}$ is the clustering-consistency contrastive loss designed to enhance the model's ability to distinguish samples with similar semantics but different classes. Further details are provided in Section 4.4. Subsequently, we employ a shared-parameter GNN encoder $\text{GNN}^{\theta}_{\text{sample}}(\cdot)$ to encode the augmented low-pass and high-pass graphs, and define the resulting sample-level node embeddings as $\mathbf{Z}^l, \mathbf{Z}^h \in \mathbb{R}^{N\times d'}$, where d' denotes the hidden dimension of the GNN encoder,. Specifically, the embeddings are encoded by:

$$\mathbf{Z}^l = \text{GNN}^{\theta}_{\text{sample}}(\tilde{\mathbf{A}}^l, \tilde{\mathbf{X}}^l), \quad \mathbf{Z}^h = \text{GNN}^{\theta}_{\text{sample}}(\tilde{\mathbf{A}}^h, \tilde{\mathbf{X}}^h). \tag{6}$$

After completing pretraining on the sample-level graphs, we further apply the encoder to the feature-level dimension graph $\mathbf{G}^{\text{dim}} = (\mathbf{A}^{\text{dim}}, \mathbf{X}^T)$. For the dimension graph, we design an independent GNN encoder $\text{GNN}^{\phi}_{\text{dim}}(\cdot)$ with parameters ϕ, which are synchronized with the sample-level encoder parameters θ during training to ensure consistency of the multi-view embedding space. The dimension node embedding, are defined as:

$$\mathbf{Z}^{\text{dim}} = \text{GNN}^{\phi}_{\text{dim}}(\mathbf{A}^{\text{dim}}, \mathbf{X}^T) \in \mathbb{R}^{d\times d'}. \tag{7}$$

Both encoders adopt Graph Convolutional Network (GCN) (Kipf & Welling, 2017) as the backbone, with the layer-wise propagation rule defined as:

$$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right), \tag{8}$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix with added self-loops, $\tilde{\mathbf{D}}$ is the corresponding degree matrix, $\mathbf{W}^{(l)}$ denotes the learnable weights at the lth layer, and $\sigma(\cdot)$ is the activation function.

4.2.2. Dimension-level enhancement

After obtaining the node embeddings from the low-pass and high-pass graphs, $\mathbf{Z}^l$ and $\mathbf{Z}^h$, we first perform an average fused embedding $\mathbf{Z}^{fuse}$ of the two views to obtain an initial combined representation:

$$\mathbf{Z}^{\text{fuse}} = \frac{1}{2}\left(\mathbf{Z}^l + \mathbf{Z}^h\right) \in \mathbb{R}^{N\times d'}. \tag{9}$$

This simple averaging effectively integrates the low- and high-frequency information while maintaining a balance between the two views (Fuchs et al., 2024). Compared to concatenation, averaging avoids dimensionality increase and additional computational overhead. Moreover, averaging promotes stable training by naturally aligning the representations from different views into the same semantic space.

Next, we apply a cross-attention mechanism (Vaswani et al., 2023) between the fused representation $\mathbf{Z}^{\text{fuse}}$ and the dimension graph embeddings $\mathbf{Z}^{\text{dim}}$, which capture the complex relationships among feature dimensions:

$$\mathbf{Z}^{\text{att}} = \text{Attention}\left(\mathbf{Z}^{\text{fuse}}, \mathbf{Z}^{\text{dim}}\right) \in \mathbb{R}^{N\times d'}. \tag{10}$$

This cross-attention enables the model to effectively capture intrinsic dependencies among feature dimensions and inject this structural information into the sample embeddings, thereby enhancing their discriminative power and expressiveness.

Finally, we again perform an average fusion between the fused sample representation $\mathbf{Z}^{\text{fuse}}$ and the attention-enhanced representation $\mathbf{Z}^{\text{att}}$ to obtain the final node embeddings:

$$\mathbf{Z}^{\text{final}} = \frac{1}{2}\left(\mathbf{Z}^{\text{fuse}} + \mathbf{Z}^{\text{att}}\right) \in \mathbb{R}^{N\times d'}. \tag{11}$$

This step smooths the integration of multi-view information, improving the stability and generalization ability of the embeddings, while avoiding dimensional redundancy and potential overfitting caused by direct concatenation.

4.3. Fine-tuning and few-shot classification

As introduced previously, we have pretrained the encoder $\mathrm{GNN}^{\theta}_{\mathrm{sample}}(\cdot)$ using multi-view contrastive learning combined with dimension-level enhancement, obtaining a generalized encoder parameter set θ. Based on this, for downstream n-way k-shot few-shot tasks, we fix the encoder parameters θ and fine-tune only a linear classifier f_{ψ}.

Specifically, for each task, we randomly select n novel classes and sample k labeled examples per class to form the support set $\mathcal{S} = \{(\mathbf{z}_i^{\mathrm{final}}, y_i)\}_{i=1}^{nk}$, and sample q unlabeled nodes to form the query set $\mathcal{Q} = \{\mathbf{z}_j^{\mathrm{final}}\}_{j=1}^{q}$.

We then optimize the linear classifier f_{ψ} on the support set by minimizing the cross-entropy loss:

$$\psi = \arg\min_{\psi} \sum_{(\mathbf{z}_i^{\mathrm{final}}, y_i)\in\mathcal{S}} \mathcal{L}_{\mathrm{CE}}\left(f_{\psi}(\mathbf{z}_i^{\mathrm{final}}), y_i\right). \tag{12}$$

After training, the classifier is evaluated on the query set $\mathcal{Q}$. This approach leverages the pretrained generalized encoder representations by freezing θ and fine-tuning only the classifier, enabling fast adaptation and stable generalization to novel classes.

4.4. Clustering-consistency contrastive loss

To train a discriminative encoder under few-shot conditions, we use high-pass and low-pass graphs as two views for contrastive learning. Embeddings of the same node across views form positive pairs, while different nodes form negative pairs. High inter-class similarity in DGA data creates hard negatives, complicating optimization. To address this, we introduce a **clustering-consistency matrix** to adaptively reweight negatives.

To capture the structural similarity between node pairs in the embedding space, we perform multiple rounds of clustering on the fusion embedding $\mathbf{Z}^{\mathrm{fuse}}$ and record the frequency with which each node pair is assigned to the same cluster. We define a consistency matrix $\mathbf{C} \in \mathbb{R}^{N\times N}$, computed as:

$$\mathbf{C}_{ij} = \frac{1}{R}\sum_{r=1}^{R} \mathbb{I}\left[\phi_r(i) = \phi_r(j)\right] \tag{13}$$

where R is the number of clustering repetitions, $\phi_r(i)$ denotes the cluster assignment of node i in the rth clustering; $\mathbb{I}[\cdot]$ is the indicator function that returns 1 if the condition holds and 0 otherwise.

To enhance clustering sensitivity to local structures, we adopt **spectral clustering** (Luxburg, 2007) to capture complex node structures in the embedding space. Based on the resulting clusters, we use the consistency matrix to reweight negative sample similarities, reducing penalties for node pairs that are structurally close in the embedding space. The contrastive loss is defined as:

$$\mathcal{L}_{\mathrm{contrast}} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp\left(\mathrm{sim}(\mathbf{z}_i^{l}, \mathbf{z}_i^{h})/\tau\right)}{\exp\left(\mathrm{sim}(\mathbf{z}_i^{l}, \mathbf{z}_i^{h})/\tau\right) + \sum_{j\neq i} w_{ij}\cdot \exp\left(\mathrm{sim}(\mathbf{z}_i^{l}, \mathbf{z}_j^{h})/\tau\right)} \tag{14}$$

where $\mathbf{z}_i^{l}, \mathbf{z}_i^{h}$ are the representations of node i under the low-pass and high-pass graphs; $\mathrm{sim}(\cdot,\cdot)$ denotes cosine similarity; τ is the temperature parameter; $w_{ij} = 1 + \alpha(1 - \mathbf{C}_{ij})$ is the consistency-based weighting factor; α controls the influence of cluster consistency.

This design effectively mitigates the impact of structurally similar but semantically different negative samples, thereby improving the encoder's discriminative ability under few-shot settings.

5. Experiments

To comprehensively evaluate the effectiveness of the proposed method for DGA, we conduct a series of experiments designed around the following three research questions:

Table 1
The Statistics Information of DGA-2000, DGA-200, and the new DGA dataset.

Dataset	PD	LE	HE	LT	MT	HT	Normal	Sum
DGA-2000	118	309	305	118	132	558	740	2321
Dataset	**SD**	**LT**	**AD**	**HT**	**MT**	**LMT**	**PD**	**Sum**
DGA-200	49	19	54	38	9	16	16	201
Dataset	**PD**	**D1**	**T1**	**D2**	**T2**	**T3**	**NF**	**Sum**
DGA-4000	351	598	463	770	349	879	740	4150

RQ1: How does the proposed method perform in terms of accuracy and stability compared to existing few-shot learning approaches under limited data conditions?

RQ2: How do the key components of the model-such as the dimension graph enhancement and the clustering consistency loss-contribute to the overall performance?

RQ3: How sensitive is the proposed method to hyperparameter settings, and which parameters have the most significant impact on performance?

RQ4: How does the proposed method perform in terms of computational efficiency, including time and memory usage, compared to other few-shot learning approaches?

5.1. Experimental setup

5.1.1. Datasets

We evaluate our approach on three public DGA datasets for DGA, The statistical information for three datasets is presents Table 1.

DGA-2000: This dataset consists of 1982 samples categorized into 7 classes: LT, MT, HT, High-Energy Discharge (HE), Low-Energy Discharge (LE), PD, and Normal. It shares the same feature dimensions as DGA-200 but suffers from class imbalance, with the largest class having 6.26 times more samples than the smallest.

DGA-200: This dataset contains 201 samples covering 7 fault types, including Partial Discharge (PD), Spark Discharge (SD), Arc Discharge (AD), Low-Temperature Overheating (LT), Low to Medium Temperature Overheating (LMT), Medium Temperature Overheating (MT), and High Temperature Overheating (HT).

DGA-4000[3]: This dataset contains 4150 samples categorized into 7 fault types: Low-Energy Discharge (D1), High-Energy Discharge (D2), Partial Discharge (PD), Thermal Faults below 300 °C (T1), Thermal Faults between 300 °C and 700 °C (T2), Thermal Faults above 700 °C (T3), and Normal (NF). It shares the same feature dimensions as DGA-200 and DGA-2000.

5.1.2. Implementation details

To simulate real-world scenarios with limited labeled data, we construct 2-way few-shot classification tasks with varying shot settings:

- On **DGA-2000**, due to the sufficient data volume and relatively balanced class distribution, we adopt 3-shot and 5-shot settings.
- On **DGA-200**, as certain classes contain too few samples to support 5-shot tasks, we use only 1-shot and 3-shot settings.
- On **DGA-4000**, we conduct experiments under 3-shot and 5-shot settings.

In each experiment, all classes are split into training, validation, and test classes in a fixed ratio of 3:2:2. For each episode, two classes from the test set are randomly selected to form a 2-way task, from which support and query samples are drawn. To ensure the reliability and reproducibility of results, we repeat each experiment with five different random seeds controlling the class splits and task sampling.

[3] https://www.kaggle.com/datasets/luizflaviopereira/dga-dissolved-gas-analysis

Table 2
Hyper-parameter settings for our method.

Parameter	DGA-2000	DGA-200	DGA-4000
n	2	2	2
k	$\{3, 5\}$	$\{1, 3\}$	$\{3, 5\}$
R	$\{1, 2, 3, 4, 5\}$	$\{1, 2, 3, 4, 5\}$	$\{1, 2, 3, 4, 5\}$
α	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$
τ	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$
Correlation threshold T	$\{0.2, 0.8\}$	$\{0.2, 0.8\}$	$\{0.2, 0.8\}$
$Epochs$	300	100	1000

In addition, we carefully tune several key hyperparameters that affect model performance. These include the number of clustering rounds R, the weight coefficient α in the contrastive loss, the temperature parameter τ, the correlation threshold T in DGE module and the total number of training epochs. The default settings are summarized in Table 2.

5.1.3. Baseline

In this work, we systematically categorize the baseline methods into two major classes to better illustrate the characteristics and advantages of each category. These two classes are GCL-based methods and Meta-learning methods.

(1) **Graph Contrastive Learning-based Methods.** These methods aim to learn generalized and robust feature representations by creating multiple augmented graph views and maximizing the consistency of node embeddings across these views. Representative methods include:
- **GRACE** (Zhu et al., 2020): GRACE constructs positive and negative sample pairs through edge dropping and feature masking, thereby improving robustness against structural and attribute perturbations.
- **BGRL** (Thakoor et al., 2023): BGRL introduces a negative-sample-free contrastive framework and leverages an online-target network with a feedback mechanism to stabilize representation learning.
- **fs-GNN** (Kipf & Welling, 2017): This method employs a pre-trained vanilla GNN encoder and utilizes few-shot tasks for downstream adaptation without additional contrastive objectives.
- **STAR** (Liu et al., 2025): This method leverages set functions and optimal transport to capture set-level features and align support-query distributions for unsupervised graph few-shot learning.

(2) **Meta-learning Methods.** Meta-learning approaches emphasize rapid model adaptation under few-shot settings, typically by simulating multi-task training to enhance generalization. Key methods include:
- **Graph Prototypical Networks (GPN)** (Ding et al., 2020): GPN extends Prototypical Networks to graphs by integrating GNNs for prototype construction and introduces an instance selection mechanism to refine prototype quality.
- **MAML** (Finn et al., 2017): Model-Agnostic Meta-Learning seeks a general initialization that can quickly adapt to new tasks with a few gradient updates, enabling efficient few-shot learning on graphs.
- **ProtoNet** (Snell et al., 2017): ProtoNet simplifies metric-based learning by computing class prototypes (centroids of support sets), allowing fast inference for few-shot classification.
- **G-Meta** (Huang & Zitnik, 2020): G-Meta introduces a meta-learning paradigm at the subgraph level to capture fine-grained structural patterns and facilitate cross-task transfer.
- **Meta-GNN** (Zhou et al., 2019): This method incorporates meta-learning strategies directly into GNN modules, enabling rapid adaptation while preserving structural information in graphs.

5.2. Performance comparison (RQ1)

As shown in Table 3, DAWN demonstrates stable and superior performance across all datasets and few-shot settings, consistently outperforming baseline methods. Key observations are summarized as follows:

Observation 1: Performance on large-scale datasets. DAWN exhibits clear advantages on larger datasets. On DGA-2000, it achieves accuracies of 0.73 (3-shot) and 0.84 (5-shot), while on DGA-4000, accuracies reach 0.72 (3-shot) and 0.80 (5-shot). Compared with other baselines, including STAR, DAWN generally performs better across settings. This indicates that, with sufficient samples, our multi-view structural modeling and unsupervised pretraining effectively capture complex graph structures and variations, enhancing representation discriminability.

Observation 2: Performance on small datasets and adaptability. Even on the smaller DGA-200 dataset with limited samples, DAWN maintains stable performance (0.63 for 1-shot) despite high uncertainty and inter-class similarity. When support samples increase to 3-shot, accuracy improves markedly to 0.70, demonstrating strong adaptability and rapid learning capability under data-scarce conditions. Other baselines show more variation and generally lower stability under the same settings.

Observation 3: Advantage scaling with support samples. The performance gap between DAWN and baseline methods widens as support samples increase. For instance, accuracy on DGA-200 rises from 0.63 (1-shot) to 0.70 (3-shot), and on DGA-2000 5-shot, DAWN outperforms all other methods with a notable margin. While some simpler baselines (e.g., ProtoNet) may achieve slight local advantages under extreme scarcity, they lack structural modeling capabilities and exhibit limited generalization.

Table 4 presents the paired t-test results for DAWN compared with the best (second-best) and worst baselines. DAWN consistently demonstrates a significant improvement over the worst baseline across all settings (t-values: 3.02–6.11, $p < 0.05$), highlighting its clear performance advantage. In contrast, compared with the second-best baseline, DAWN achieves slight improvements in most cases, but the differences are not statistically significant (t-values: −0.09–1.43, $p \geq 0.05$), indicating that its performance is comparable to the strongest competitor while maintaining robustness.

Overall, DAWN achieves more stable and superior average performance across all datasets. These results further confirm that our multi-view structural modeling leverages complementary graph perspectives, while contrastive learning enhances embedding discriminability. As support samples increase, DAWN extracts features more efficiently, achieving stronger discriminative power, transferability, and sample efficiency.

5.3. Study of DAWN (RQ2)

To evaluate the effectiveness of DWAN's core components—**Clustering-consistency Contrastive Learning (CCL)** and **Dimension Graph Enhancement (DGE)**—we conduct three experiments addressing RQ2: (1) Ablation Study, where we remove the DGE and CCL modules to assess their impact on performance; (2) CCL Module Analysis, where we explore various class division strategies to examine their effects on generalization and accuracy improvements. and (3) DGE Module Analysis, in which we compare different dimension graph construction methods and integrate DGE into baseline models to validate its feature enhancement capability;

5.3.1. Ablation studies

To assess the individual contributions of key components in our model, we conduct ablation studies on the DGA-2000 and DGA-200 datasets under various few-shot classification settings. Specifically, to

Table 3
Few-shot classification accuracy on DGA-200, DGA-2000, and the new DGA dataset. Each result is reported as mean ± standard deviation. Results highlighted in **bold** indicate the best performance, while those highlighted in italic indicate the second-best performance.

Datasets	DGA-2000		DGA-200		DGA-4000	
Task	2-way 3-shot	2-way 5-shot	2-way 1-shot	2-way 3-shot	2-way 3-shot	2-way 5-shot
GCL-based Methods						
GRACE	0.65 ± 0.13	0.63 ± 0.12	0.61 ± 0.17	0.67 ± 0.14	0.60 ± 0.12	0.64 ± 0.11
BGRL	0.65 ± 0.12	0.67 ± 0.10	0.61 ± 0.15	0.69 ± 0.15	0.63 ± 0.11	0.63 ± 0.11
Fs-GNN	0.59 ± 0.10	0.62 ± 0.10	*0.63 ± 0.16*	0.67 ± 0.14	0.55 ± 0.07	0.58 ± 0.09
STAR	*0.71 ± 0.16*	*0.84 ± 0.16*	0.60 ± 0.16	**0.75 ± 0.21**	*0.70 ± 0.15*	*0.79 ± 0.13*
Meta-learning Methods						
GPN	0.62 ± 0.06	0.71 ± 0.09	0.55 ± 0.14	0.60 ± 0.15	0.64 ± 0.04	0.67 ± 0.04
MAML	0.68 ± 0.14	0.67 ± 0.13	0.62 ± 0.17	0.61 ± 0.15	0.70 ± 0.17	0.69 ± 0.17
ProtoNet	0.66 ± 0.13	0.58 ± 0.12	0.63 ± 0.18	0.58 ± 0.15	0.70 ± 0.16	0.63 ± 0.15
G-Meta	0.52 ± 0.09	0.50 ± 0.04	0.50 ± 0.03	0.50 ± 0.01	0.56 ± 0.11	0.52 ± 0.07
Meta-GNN	0.55 ± 0.11	0.53 ± 0.08	0.53 ± 0.09	0.50 ± 0.02	0.53 ± 0.11	0.56 ± 0.14
DAWN (Ours)	**0.83 ± 0.10**	**0.84 ± 0.09**	**0.64 ± 0.15**	*0.70 ± 0.13*	**0.74 ± 0.11**	**0.80 ± 0.10**

Table 4
Paired t-test results for DAWN compared with best and worst baselines on DGA datasets.

Dataset	Setting	vs Best Baseline	vs Worst Baseline
DGA-2000	2-way 3-shot	$t = 1.43$, $p = 0.20$ (STAR)	$t = 6.11$, $p = 0.00^*$ (G-Meta)
	2-way 5-shot	$t = -0.09$, $p = 0.93$ (STAR)	$t = 3.39$, $p = 0.01^*$ (G-Meta)
DGA-200	2-way 1-shot	$t = 0.28$, $p = 0.79$ (fsGNN)	$t = 3.61$, $p = 0.02^*$ (G-Meta)
	2-way 3-shot	$t = 0.22$, $p = 0.83$ (STAR)	$t = 3.02$, $p = 0.04^*$ (G-Meta)
DGA-4000	2-way 3-shot	$t = 0.41$, $p = 0.69$ (STAR)	$t = 3.25$, $p = 0.03^*$ (Meta-GNN)
	2-way 5-shot	$t = 0.14$, $p = 0.89$ (STAR)	$t = 4.60$, $p = 0.01^*$ (G-Meta)

t and p values are shown, * indicates significance ($p < 0.05$), baseline names are indicated in parentheses.

Table 5
Ablation Study Results (Accuracy ± Std) on DGA datasets.

Datasets	DGA-2000		DGA-200		DGA-4000	
	2-way 3-shot	2-way 5-shot	2-way 1-shot	2-way 3-shot	2-way 3-shot	2-way 5-shot
w/o CCL	0.68 ± 0.10	0.72 ± 0.10	0.62 ± 0.16	0.67 ± 0.15	0.68 ± 0.12	0.69 ± 0.09
w/o DGE	0.67 ± 0.12	0.69 ± 0.11	0.60 ± 0.16	0.66 ± 0.13	0.61 ± 0.11	0.62 ± 0.11
w/o DGE & CCL	0.65 ± 0.11	0.66 ± 0.10	0.60 ± 0.15	0.65 ± 0.16	0.60 ± 0.11	0.62 ± 0.11
DAWN (Ours)	**0.83 ± 0.10**	**0.84 ± 0.09**	**0.64 ± 0.15**	**0.70 ± 0.13**	**0.74 ± 0.11**	**0.80 ± 0.10**

assess the impact of our proposed components, we perform ablation studies by removing the and **CCL** the **DGE** modules individually and observe the resulting performance changes.

We perform the following ablation variants:

- **w/o CCL**: removes the clustering consistency loss; model is trained using a standard InfoNCE objective.
- **w/o DGE**: removes the dimensional graph module; contrastive learning is performed only on raw sample features.
- **w/o DGE & CCL**: removes both DGE and CCL, serving as a fully ablated baseline.

The ablation results in Table 5 highlight the significant and complementary contributions of both the DGE and CCL modules. We have the following observations:

Observation 1: Essential synergy. The full model combining DGE and CCL achieves optimal performance across datasets (e.g., **0.84** on DGA-2000 and **0.80** on DGA-4000, 5-shot). Removing both modules causes a substantial drop (e.g., **−0.18** to **0.66** on DGA-2000 and **−0.18** to **0.62** on DGA-4000, 5-shot), demonstrating their strong synergistic effect in enhancing representational power.

Observation 2: Consistent gains in low-resource settings. Both modules consistently improve performance under few-shot conditions. On DGA-200 (3-shot), accuracy drops significantly upon removing either module (from **0.70** to **0.66** w/o DGE, and to **0.67** w/o CCL), and on DGA-4000 (3-shot), removing DGE or CCL reduces accuracy from **0.72** to **0.61**/**0.68**, confirming their robustness and effectiveness even with limited data.

Observation 3: DGE's advantage in larger-data settings. The relative importance of DGE increases with more data. On DGA-2000 (5-shot), removing DGE causes a larger performance drop (**−0.15** to **0.69**) than removing CCL (**−0.12** to **0.72**). This trend is also observed on DGA-4000 (5-shot), where removing DGE reduces accuracy from **0.80** to **0.62**, while removing CCL reduces it to **0.69**, suggesting DGE's inter-dimensional relationship modeling becomes particularly crucial when richer structural cues are available.

Observation 4: Limited standalone impact in extreme low-shot. In the most challenging DGA-200 (1-shot) setting, removing either module individually yields similar, limited degradation (to **0.60** w/o DGE, **0.62** w/o CCL), indicating that under extreme data scarcity, the standalone contribution of each module is constrained, highlighting the difficulty of learning in this regime.

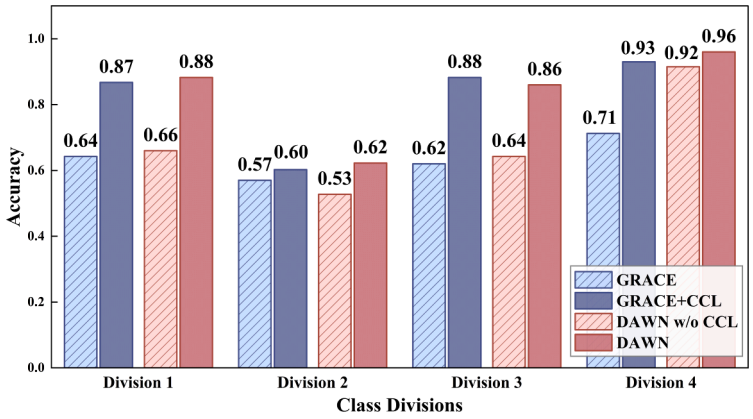


Fig. 6. Classification accuracy comparison of baseline GRACE, e, GRACE enhanced with CCL (GRACE + CCL), DAWN w/o CCL, and full DAWN model across different class divisions under the 2-way 5-shot setting on the DGA-2000.

Table 6
Details of class divisions with corresponding training and testing classes and their average inter-class similarities.

Class Division	Train Classes	Train Sim	Test Classes	Test Sim
Division 1	Normal, HE, LT	0.517	LE, HT	0.46
Division 2	HE, LE, HE	0.60	MT, HT	0.65
Division 3	LE, MT, PD	0.56	Normal, LE	0.43
Division 4	PD, LT, HT	0.52	Normal, LE	0.43

Overall, DGE and CCL provide stable, complementary benefits across datasets. Their joint integration is most effective, especially as support samples increase. However, their individual impact is more modest in extreme 1-shot scenarios, suggesting potential for future enhancements.

5.3.2. *Effectiveness of the CCL module*

To evaluate the efficacy of the CCL module in mitigating challenges posed by high inter-class similarity, we conducted 2-way 5-shot experiments on the DGA-2000 dataset. Four distinct class divisions representing varying task difficulties are utilized, as detailed in Table 6. Fig. 6 compares the classification accuracies achieved by four models: the baseline GRACE, GRACE enhanced with CCL (GRACE + CCL), DAWN without CCL (DAWN w/o CCL), and the full DAWN model. Key findings from these experiments are as follows:

First, we observe the significant Impact of inter-class similarity on model performance. On the training set level, higher training set inter-class similarity (e.g., Divisions 2 and 3 in) leads to weaker generalization ability. For instance, the GRACE model's accuracy drops by 9% in Division 3 compared to Division 4, which has lower training set similarity. On the testing set level, higher testing set inter-class similarity (e.g., Divisions 1 and 2 in) increases classification confusion. For example, although Division 1 has training set similarity comparable to Division 4, its classification accuracy is notably lower.

Secondly, regarding the CCL Module Effectiveness, we have the following observations:

Observation 1: CCL has the ability to enhance generalization power. The CCL significantly improves the model's adaptability under high training set similarity conditions. In Division 3, which exhibits high training similarity, the full DAWN model achieves the highest accuracy (0.86), outperforming both the DAWN variant without CCL (0.64) and the baseline GRACE model (0.71). Notably, in Division 2, characterized by the highest training similarity, the DAWN model achieves the best classification accuracy (0.62).

Observation 2: CCL improves the discriminative power, enabling the model to better resist confusion induced by high testing set similarity. In divisions with high testing set similarity (Divisions 1 and 2), the DAWN model achieves the highest classification accuracies (0.88 and 0.62, respectively) and notably outperforms all other model variants, effectively reducing classification confusion.

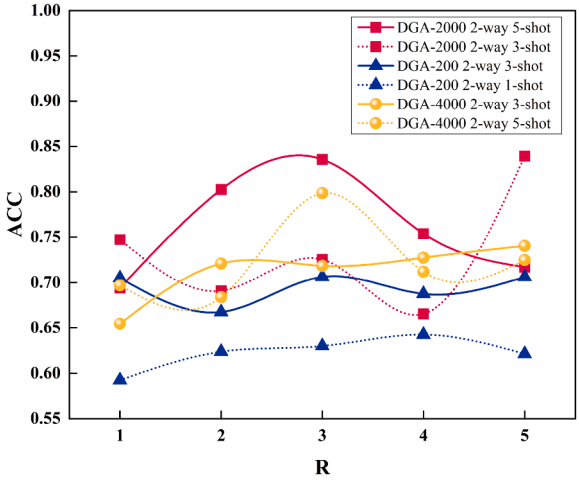
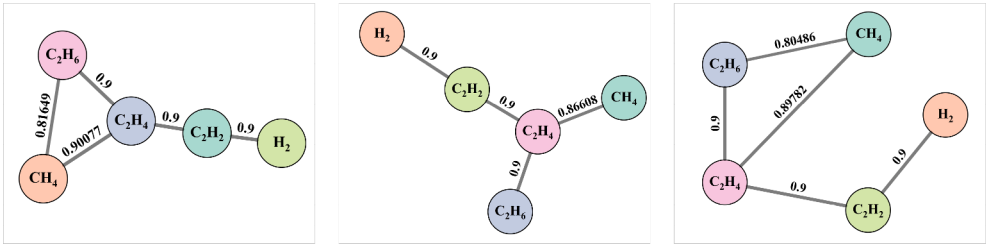


Fig. 7. Classification accuracy comparison of baseline GRACE, e, GRACE enhanced with CCL (GRACE + CCL), DAWN w/o CCL, and full DAWN model across different class divisions under the 2-way 5-shot setting on the DGA-2000.

To further analyze the behavior of CCL under different clustering configurations, we explore the impact of the clustering rounds R.

As shown in Fig. 7, we examine how the number of clustering rounds R affects model performance across different datasets. A consistent "rise-then-fall" pattern is observed as R increases from 1 to 5. When R is small, the clustering process cannot adequately capture the structural consistency among representations, limiting the strength of the consistency constraint. Moderate rounds (e.g., $R = 3$) enable a better balance between capturing local consistency and avoiding over-clustering, thus yielding the best results on most datasets. However, excessive clustering rounds may introduce redundant or unstable grouping information, reducing model stability and generalization. These findings further verify that the CCL module functions most effectively under a proper clustering configuration, ensuring stable and discriminative representations.

In summary, the CCL module effectively enhances model performance by addressing two key challenges: (1) weakened generalization caused by high training-set similarity, and (2) classification confusion arising from high testing-set similarity. Together with the analysis of clustering rounds R, these results demonstrate that CCL can produce



(a) Dimension graph of DGA-2000 (b) Dimension graph of DGA-200 (c) Dimension graph of DGA-4000

Fig. 8. Topology of the dimension graphs across three datasets.

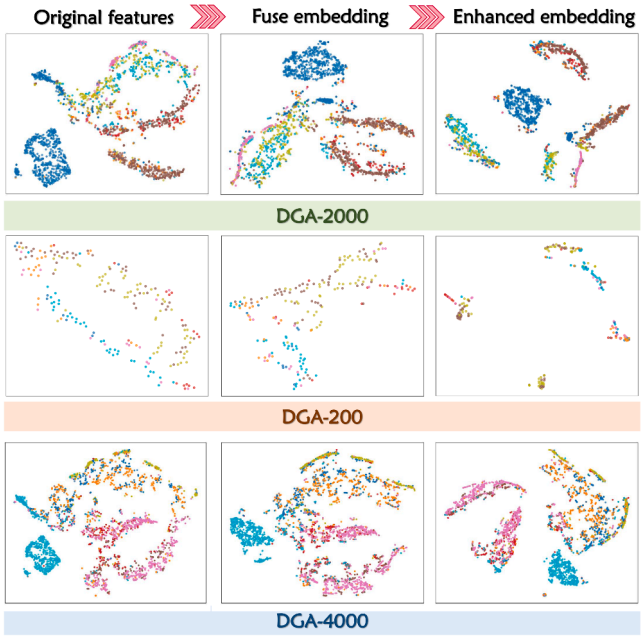


Fig. 9. T SNE visualization of feature embeddings on all datasets: (a) original features, (b) fused embeddings (w/o DGE), (c) enhanced embeddings (w/ DGE).

stable and discriminative representations under appropriate configurations, thereby improving reliability in few-shot fault diagnosis tasks.

5.3.3. *Effectiveness of the DGE module*

This subsection begins by presenting the dimension graphs for each dataset (Fig. 8), constructed with a correlation threshold of 0.8 to visually represent the gas relationships. We then analyze the corresponding t-SNE embeddings (Fig. 9), which compare the original, fused, and enhanced feature spaces. The visualizations clearly demonstrate that the DGE module significantly improves intra-class compactness and inter-class separation, confirming its efficacy in enhancing feature discriminability.

We further evaluate the role of the DGE module by designing three variants of the GRACE baseline: using only the dimension graph built on prior knowledge (Prior), using only the correlation-based dimension graph (Corr), and combining both (DGE). Table 7 presents the performance of these variants on different datasets, and the results reveal the following insights:

Observation 1: Complementary advantages. Using Corr alone captures latent statistical relationships, improving the 5-shot accuracy on DGA-2000 from 0.63 to 0.72, whereas using Prior alone further boosts it to 0.78. This indicates that domain knowledge provides additional discriminative information that complements statistical correlations, significantly enhancing the model's ability to distinguish between classes. On the larger DGA-4000 dataset, Prior yields the largest gain (5-shot 0.71), while Corr alone results in moderate improvement (5-shot 0.68), suggesting that domain knowledge is particularly important for large-scale data and plays a key role in overall performance improvement. These results highlight the complementary nature of statistical and domain-driven information in few-shot fault diagnosis.

Observation 2: Consistent performance improvement with DGE. Combining Prior and Corr to construct the DGE module achieves the best performance across most datasets and settings. On DGA-2000 (5-shot), DGE raises GRACE accuracy from 0.63 to 0.80 and DAWN accuracy from 0.69 to 0.84, demonstrating substantial gains over the individual components. On DGA-4000, DGE also produces significant improvement, increasing DAWN 5-shot accuracy from 0.62 (without DGE) to 0.80. Even on the smaller DGA-200 dataset, DGE maintains stable improvement (0.69 vs. 0.66), confirming its robustness and scalability across datasets of varying sizes. Overall, these results indicate that DGE effectively integrates statistical correlations and domain knowledge, leveraging complementary and synergistic effects to consistently enhance model performance in few-shot scenarios.

Fine-grained analysis of gas relationships. To examine the contribution of individual gas relationships within the DGE module, we conduct ablation studies as shown in Table 8, where each experiment removes one pairwise gas relation while keeping other components unchanged. The results show that removing the C2H4–C2H2 relation causes the largest performance drop, highlighting its critical role in fault identification, whereas removing the C2H6–C2H4 relation has a minor impact, indicating its limited contribution. Other gas relations show varying importance across datasets, but collectively, their interactions in the dimension graph significantly enhance the model's discriminative ability and stability. Analysis of Tables 7 and 8 indicates that the effectiveness of DGE relies on the integration of domain knowledge, statistical correlations, and the synergistic effect of multiple gas relationships.

Overall, the DGE module consistently and significantly improves the performance of GRACE and DAWN, demonstrating that constructing dimension graphs integrating domain-specific priors and statistical correlations can effectively enhance discriminative power and generalization capability in few-shot fault diagnosis, while ensuring stable performance across datasets of varying sizes and configurations.

5.4. *Hyperparameter analysis (RQ3)*

In this section, we investigate the influence of key hyperparameters on the performance and stability of the proposed model. The analysis is divided into three parts: (1) Impact of the number of labeled samples per class (k-shot) on model performance; (2) the influence of the Weight

Table 7
Comparison of the Dimension Graph Enhancement module (DGE) and its components—prior-based (Prior) and correlation-based (Corr) graph construction—on classification accuracy of the GRACE baseline and DAWN framework under various few-shot settings on all datasets.

Method	Variants	Prior	Corr	DGA-2000		DGA-200		DGA-4000	
				2-way 3-shot	2-way 5-shot	2-way 1-shot	2-way 3-shot	2-way 3-shot	2-way 5-shot
GRACE	GRACE	✗	✗	0.65 ± 0.13	0.63 ± 0.12	0.61 ± 0.17	0.67 ± 0.14	0.60 ± 0.12	0.64 ± 0.11
	+ Prior	✓	✗	0.67 ± 0.10	0.78 ± 0.07	0.62 ± 0.16	0.67 ± 0.13	**0.70 ± 0.10**	**0.71 ± 0.08**
	+ Corr	✗	✓	0.71 ± 0.11	0.72 ± 0.09	0.61 ± 0.15	0.67 ± 0.13	0.67 ± 0.12	0.68 ± 0.09
	+ DGE	✓	✓	**0.72 ± 0.08**	**0.80 ± 0.09**	**0.64 ± 0.16**	**0.69 ± 0.14**	0.70 ± 0.13	0.69 ± 0.07
DAWN (Ours)	w/o DGE	✗	✗	0.67 ± 0.12	0.69 ± 0.11	0.66 ± 0.13	0.66 ± 0.13	0.61 ± 0.11	0.62 ± 0.11
	w/o Corr	✓	✗	0.71 ± 0.10	0.83 ± 0.06	0.62 ± 0.15	0.66 ± 0.16	0.72 ± 0.11	0.75 ± 0.10
	w/o Prior	✗	✓	0.71 ± 0.10	0.79 ± 0.07	0.58 ± 0.15	0.63 ± 0.15	0.71 ± 0.12	0.71 ± 0.10
	DAWN	✓	✓	**0.83 ± 0.10**	**0.84 ± 0.09**	**0.64 ± 0.15**	**0.70 ± 0.13**	**0.74 ± 0.11**	**0.80 ± 0.10**

Table 8
Ablation Study Results (Accuracy ± Std).

Fault Type	DGA-2000		DGA-200		DGA-4000	
	2-way 3-shot	2-way 5-shot	2-way 1-shot	2-way 3-shot	2-way 3-shot	2-way 5-shot
w/o $H_2 \leftrightarrow C_2H_2$	0.71 ± 0.10	0.73 ± 0.09	0.61 ± 0.13	0.70 ± 0.12	0.67 ± 0.12	0.72 ± 0.10
w/o $C_2H_6 \leftrightarrow C_2H_4$	0.70 ± 0.09	0.78 ± 0.11	0.64 ± 0.17	0.67 ± 0.15	0.70 ± 0.12	0.77 ± 0.08
w/o $C_2H_4 \leftrightarrow C_2H_2$	0.69 ± 0.09	0.67 ± 0.09	0.62 ± 0.16	0.68 ± 0.12	0.71 ± 0.10	0.69 ± 0.10
DAWN (Ours)	**0.83 ± 0.10**	**0.84 ± 0.09**	**0.64 ± 0.15**	**0.70 ± 0.13**	**0.74 ± 0.11**	**0.80 ± 0.10**

Table 9
Effect of Correlation Threshold T (Accuracy ± Std).

Threshold T	DGA-2000		DGA-2000 (Low-Mem)		DGA-4000	
	2-way 3-shot	2-way 5-shot	2-way 1-shot	2-way 3-shot	2-way 3-shot	2-way 5-shot
$T = 0.2$	0.79 ± 0.10	0.73 ± 0.10	0.63 ± 0.16	0.67 ± 0.15	0.74 ± 0.11	0.76 ± 0.08
$T = 0.8$	**0.84 ± 0.10**	**0.84 ± 0.08**	**0.64 ± 0.15**	**0.71 ± 0.16**	**0.74 ± 0.11**	**0.80 ± 0.10**

Coefficient α and Temperature Parameter τ in CCL module; and (3) the effect of the edge threshold in the DGE module.

(1) **Impact of the number of labeled samples per class (k-shot) on model performance.** Under different k-shot settings, we evaluated the model's performance on each fault class (Fig. 10). For DGA-2000, the accuracy increases from 0.74 at $k = 1$ to 0.83 at $k = 3$, and slightly improves to 0.84 at $k = 5$. The trends for DGA-4000 and DGA-200 are similar, reaching 0.80 and 0.78, respectively. Overall, performance improves significantly when the number of labeled samples per fault class increases from 1 to 3, while further gains from 3 to 5 are modest. These results indicate that the model can achieve effective fault recognition even with very few labeled samples, and performance further improves as more samples are available. This analysis also provides guidance for industrial deployment: reliable online monitoring can be achieved under extremely limited labeled data, while performance can be further optimized by moderately increasing the number of samples.

(2) **Influence of the Weight Coefficient α and Temperature Parameter τ in the CCL Module.** Fig. 11(a) and (b) show the model performance on DGA-2000 under the 2-way 5-shot setting, with all other parameters fixed. The weight coefficient α controls the balance between the contrastive consistency loss and the main classification objective. Small α values stabilize training by letting the main task dominate, whereas overly large α can cause performance fluctuations. The temperature parameter τ adjusts the sharpness of the similarity distribution in contrastive learning: too small τ increases gradient variance, while too large τ reduces the distinction between positive and negative pairs. Moderate settings ($\alpha = 0.1-0.5$, $\tau \approx 0.5$) achieve a good trade-off between stability and discriminative ability, yielding the highest accuracy.

(3) **Influence of the Correlation Threshold T in DGE module.** The threshold T controls the connectivity of the dimension graph, where an edge is formed between two dimensions only if their correlation exceeds T. The correlation values are distributed within the range $[0.1, 0.9]$, so we select two representative thresholds: $T = 0.2$ corresponds to a low threshold, resulting in a dense graph with more weakly correlated edges and richer topology; $T = 0.8$ corresponds to a high threshold, keeping only strongly correlated dimensions and yielding a sparser graph. As shown in Table 9, across all few-shot settings, $T = 0.8$ generally achieves better performance than $T = 0.2$, indicating that retaining highly correlated dimensions enhances the discriminative power of the representations. Meanwhile, $T = 0.2$ provides more global structural information despite slightly lower ac-

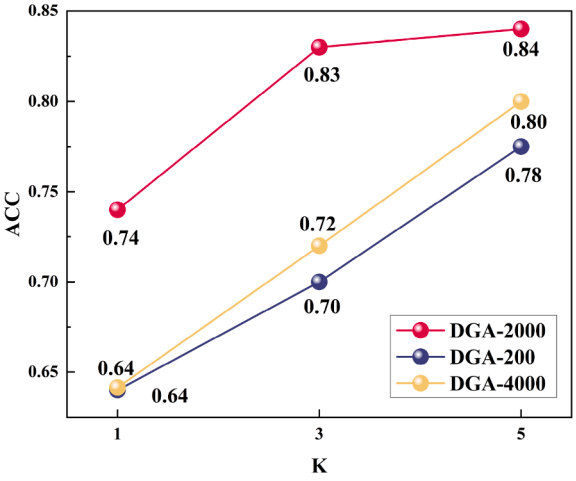


Fig. 10. Accuracy of the model under different k-shot settings for three datasets, respectively, showing performance trends with increasing labeled samples.

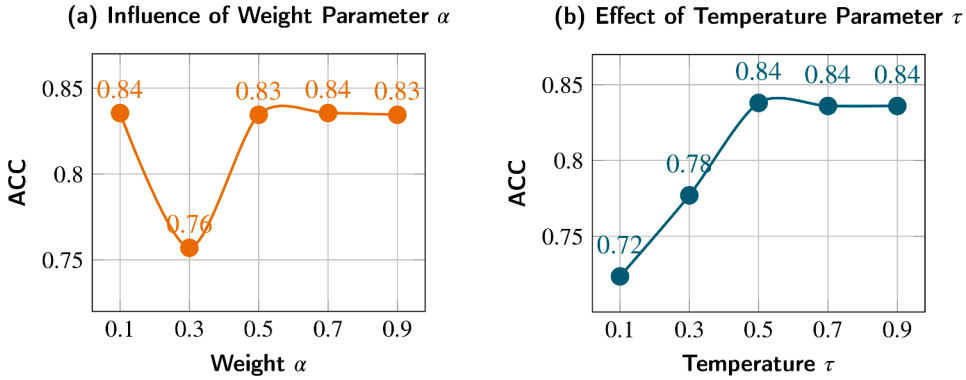


Fig. 11. Hyperparameter sensitivity analysis for our model on DGA-2000 2-way 5-shot setting,with all other parameters fixed. (a) presents the impact of the weight parameter α in adjusting negative sample importance, with $\alpha = 0.1$ and $\alpha = 0.7$ yielding the highest accuracy. (b) illustrates the effect of temperature τ in the contrastive loss; a moderate temperature ($\tau = 0.5$) gives the best result.

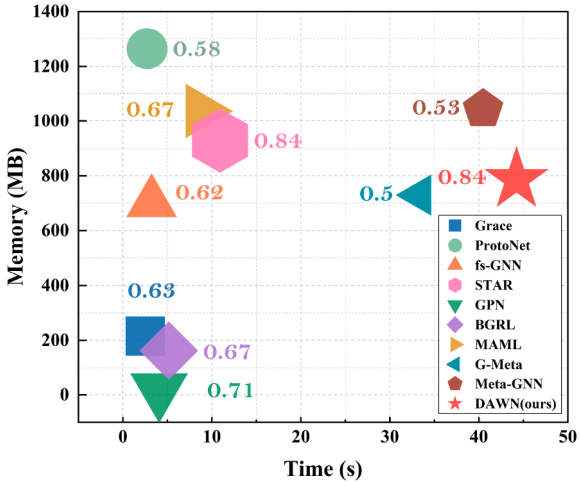


Fig. 12. Comparison of time and memory consumption among different methods on the DGA2000 dataset under the 2-way 5-shot setting. The marker size denotes the classification accuracy.

curacy. These results suggest that properly balancing graph sparsity and information richness is crucial for obtaining high-quality dimension representations.

Overall, few-shot performance improves with more labeled samples, moderate α and τ ensure CCL stability and discriminability, and a higher DGE threshold T preserves key dimension relations, highlighting the importance of proper hyperparameter selection.

5.5. Time and space complexity analysis (RQ4)

We further analyze the computational efficiency of the proposed model in terms of time and space complexity. For a graph with N nodes, E edges, and feature dimension d, the encoder requires $O(E + Nd)$ operations, while the multi-view contrastive module introduces an additional $O(Nd)$ cost due to representation alignment across high- and low-pass views. The dimension graph contributes $O(d^2)$ complexity, but since $d \ll N$, the overall computational cost remains dominated by the encoder. Memory usage primarily comes from the embedding storage and intermediate contrastive projections, resulting in approximately 700 MB consumption on the DGA2000 dataset under the 2-way 5-shot setting. As shown in Fig. 12, the proposed model maintains a reasonable time–memory trade-off compared to baselines, while achieving competitive accuracy. These results demonstrate that the overall computation remains within a controllable range, making the model practical for real-world applications.

6. Conclusion

This paper addresses the challenges of few-shot learning in transformer fault diagnosis using DGA by proposing the novel DAWN framework. DAWN integrates a contrastive few-shot learning module with clustering consistency loss to enhance the discriminability of highly similar fault categories through lightweight fine-tuning. Additionally, it incorporates a knowledge-enhanced dimension graph that explicitly models complex structural dependencies among gas features by combining statistical correlations with expert domain knowledge, enriching sample representations. Extensive experiments on benchmark datasets demonstrate that DAWN significantly outperforms traditional methods and existing deep learning models in few-shot classification tasks, with particularly notable gains in rare fault detection. As the first contrastive few-shot learning framework specifically tailored for DGA, DAWN not only achieves state-of-the-art performance but also provides new insights and a solid foundation for future research on integrating data-driven and knowledge-driven graph learning in fault diagnosis.

CRediT authorship contribution statement

Jiyuan Sun: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration; **Huifang Ma:** Conceptualization, Methodology, Software, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition; **Shuai Yang:** Conceptualization, Software, Investigation, Supervision, Writing – original draft, Writing – review & editing; **Rui Bing:** Software, Investigation, Supervision, Writing – original draft, Writing – review & editing; **Zhixin Li:** Software, Investigation, Supervision, Writing – original draft, Writing – review & editing.

Data availability

I provided the dataset source in the main text.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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